

EDA

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Clean the Data

```
library(dplyr)
library(tidyverse)
library(ggplot2)
library(tsibble)
library(feasts)
library(fable)
```

```
singapore <- read.csv('BirthsAndFertilityRatesAnnual.csv')
```

```
clean_1 <- singapore |>
  filter(DataSeries %in% c("Total Fertility Rate (TFR)", "Total Live-Births"))
```

```
clean_2 <- pivot_longer(clean_1,
  cols = starts_with("X"),
  names_to = "Year",
  values_to = "Value",
  values_transform = list(Value = as.character))
```

```
clean_3 <- mutate(clean_2, Year = str_remove(Year, "X"))
```

```
clean_4 <- clean_3 |> mutate(Year = as.numeric(Year), Value = as.numeric(Value))
```

```
clean_data <- clean_4 |>
  pivot_wider(names_from = DataSeries, values_from = Value)
singapore <- clean_data |>
  as_tsibble(index = Year)
singapore
```

```
## # A tsibble: 66 x 3 [1Y]
##   Year 'Total Fertility Rate (TFR)' 'Total Live-Births'
##   <dbl> <dbl> <dbl>
## 1 1960 5.76 61775
## 2 1961 5.41 59930
## 3 1962 5.21 58977
## 4 1963 5.16 59530
## 5 1964 4.97 58217
```

```
## 6 1965          4.66          55725
## 7 1966          4.46          54680
## 8 1967          3.91          50560
## 9 1968          3.53          47241
## 10 1969         3.22          44562
## # i 56 more rows
```

The data has been extracted, and the EDA is only on the Total Fertility Rate (TFR) and Total Live-Births (TLB).

```
singapore |> autoplot(`Total Fertility Rate (TFR)` ) +
  labs(title = "Total Fertility Rate (1960-2024)",
        caption = "Figure 1. Plot of the Total Fertility Rate in Singapore from 1960 to 2024.",
        y = "Total Fertility Rate",
        x = "Year") +
  theme(plot.caption = element_text(hjust = 0.5),
        plot.title = element_text(hjust = 0.5))
```

```
## Warning: 'autoplot.tbl_ts()' was deprecated in fabletools 0.6.0.
## i Please use 'ggtime::autoplot.tbl_ts()' instead.
## i Graphics functions have been moved to the {ggtime} package. Please use
##   'library(ggtime)' instead.
## This warning is displayed once per session.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

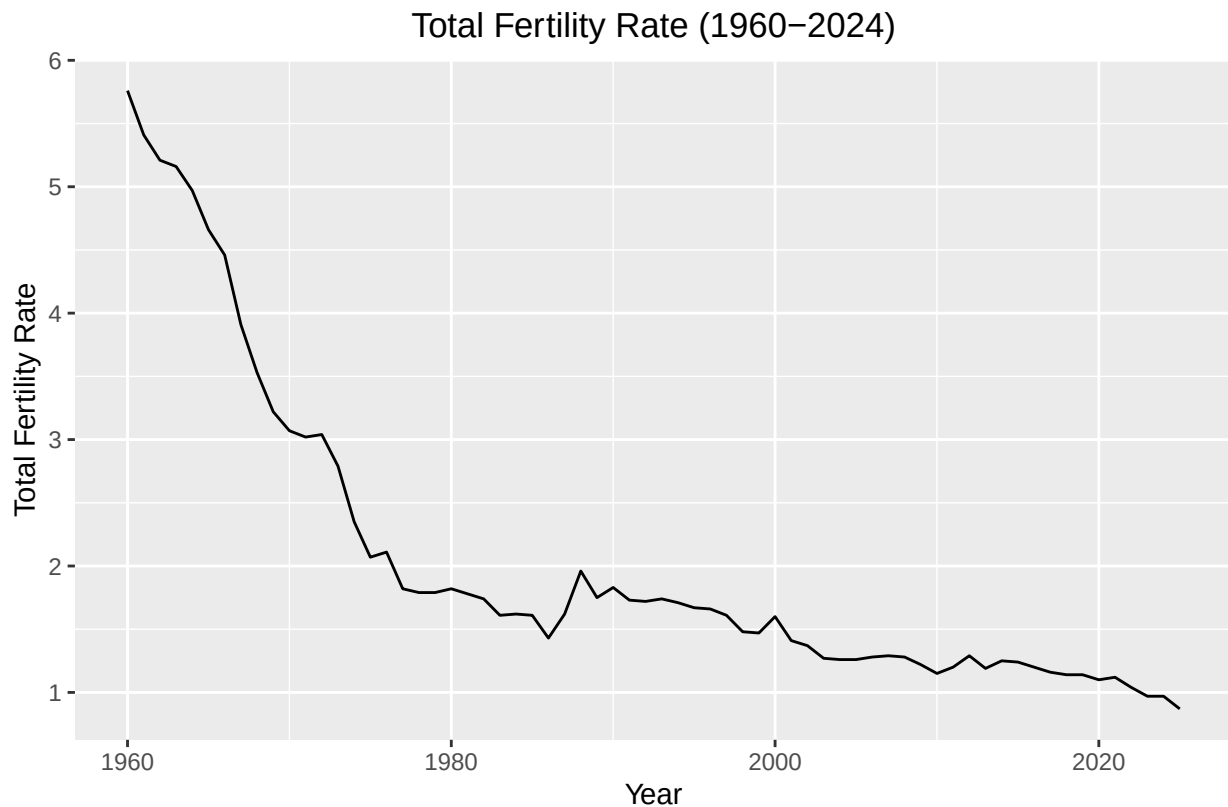


Figure 1. Plot of the Total Fertility Rate in Singapore from 1960 to 2024.

```
singapore |>
  autoplot(`Total Live-Births`) +
  labs(
    title = "Total Live-Births (1960-2024)",
    caption = stringr::str_wrap(
      paste0(
        "Figure 2. Plot of the Total Live-Births in Singapore from",
        " 1960 to 2024 shows that it follows a roughly similar trend",
        " as the plot of the Total Fertility Rate."
      ),
    ),
    width = 80
  ),
  y = "Total Live-Births",
  x = "Year") +
  theme(plot.caption = element_text(hjust = 0.5),
        plot.title = element_text(hjust = 0.5))
```

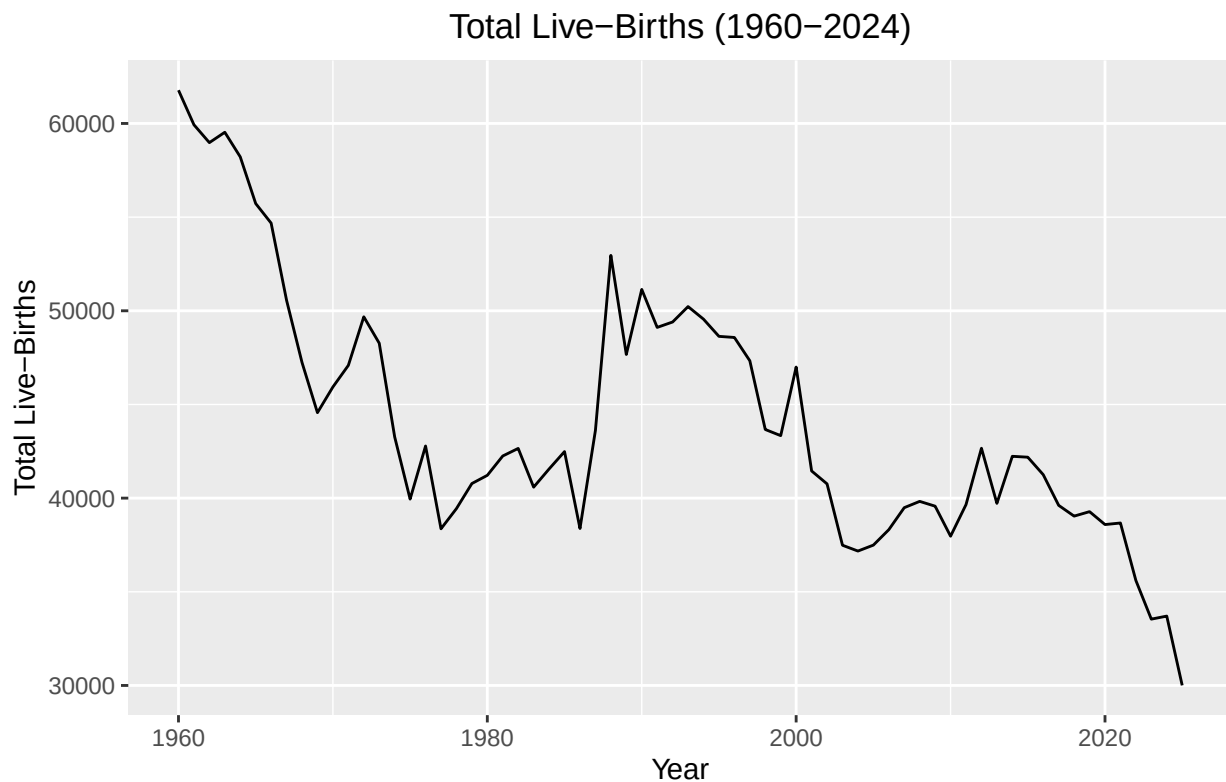


Figure 2. Plot of the Total Live-Births in Singapore from 1960 to 2024 shows that it follows a roughly similar trend as the plot of the Total Fertility Rate.

While figure 1 and figure 2 look different we can see that they show spikes and drops at similar points. There is a spike around 1988 in the plot of Total Fertility Rate and a large spike around the same year in the plot of the Total Live-Births. This can also be seen with a spike around 1972, as well as drops around 1969, 1987, 2003, and 2024.

```
log_TFR <- log(singapore$`Total Fertility Rate (TFR)`)
log_TLB <- log(singapore$`Total Live-Births`)
```

```
singapore |> autoplot(log_TFR) +
  labs(title = "Log-Transformed Total Fertility Rate (1960-2024)",
        caption = stringr::str_wrap("Figure 3. Plot of the Total Fertility Rate with a log transformation in Singapore from 1960 to 2024."),
        y = "log(Total Fertility Rate)",
        x = "Year") +
  theme(plot.caption = element_text(hjust = 0.5),
        plot.title = element_text(hjust = 0.5))
```

```
## Ignoring unknown labels:
## * width : "80"
```

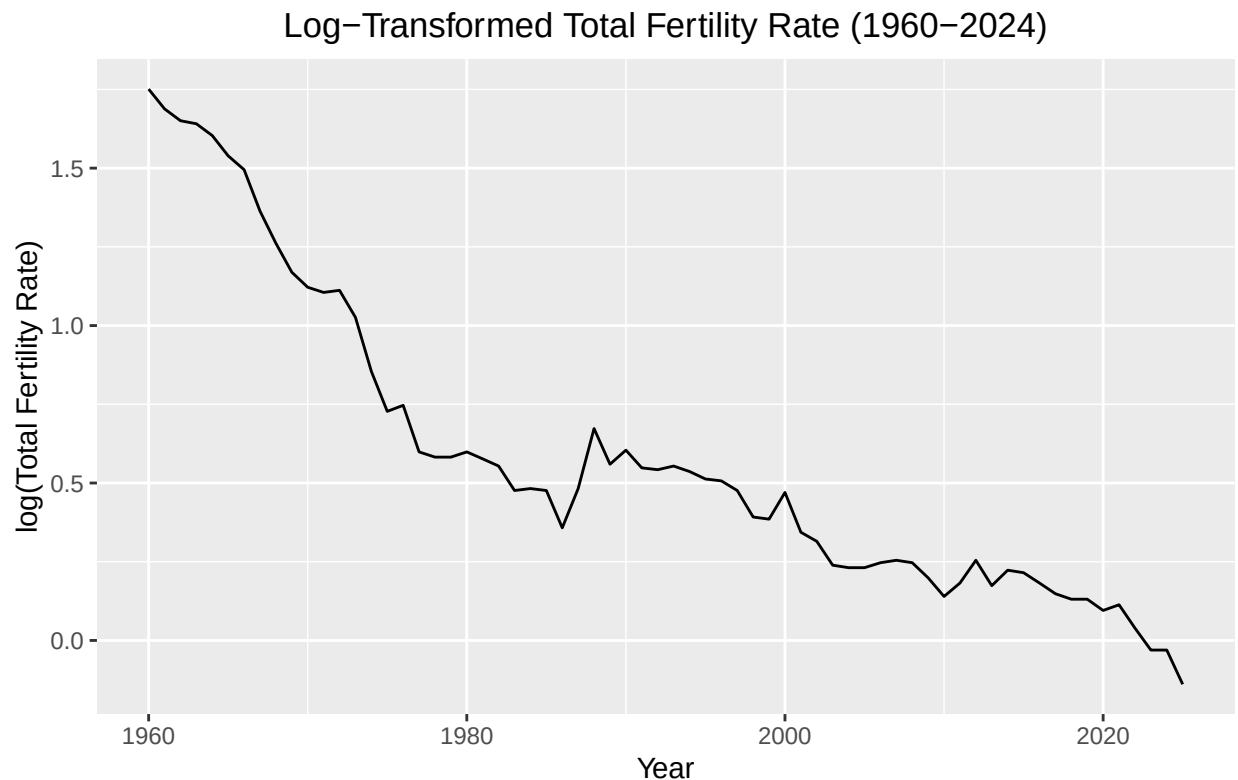


Figure 3. Plot of the Total Fertility Rate with a log transformation in Singapore from 1960 to 2024.

```
singapore |> autoplot(log_TLB) +
  labs(title = "Log-Transformed Total Live-Births (1960-2024)",
        caption = stringr::str_wrap("Figure 4. Plot of the Total Live-Births with a log transformation in Singapore from 1960 to 2024."),
        y = "log(Total Live-Births)",
        x = "Year") +
  theme(plot.caption = element_text(hjust = 0.5),
        plot.title = element_text(hjust = 0.5))
```

```
## Ignoring unknown labels:
## * width : "80"
```

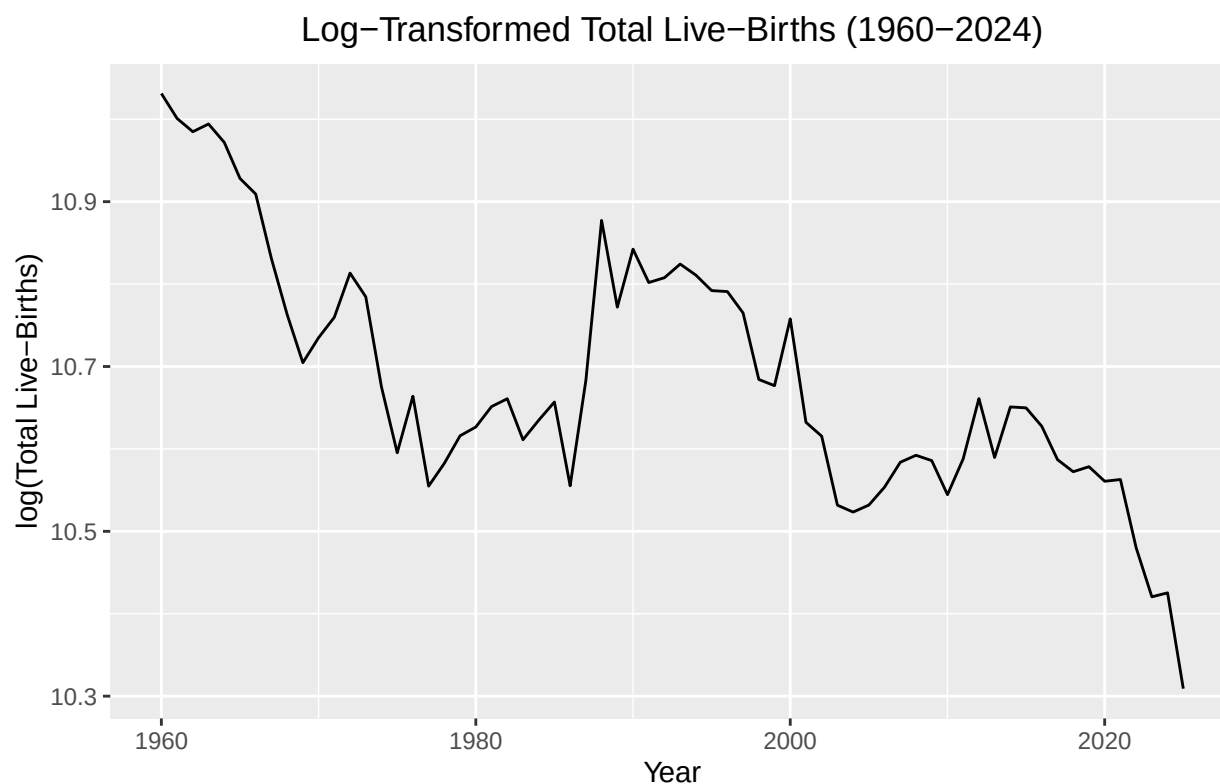


Figure 4. Plot of the Total Live-Births with a log transformation in Singapore from 1960 to 2024.

```
singapore |> features(log_TFR, unitroot_kpss)
```

```
## # A tibble: 1 x 2
##   kpss_stat kpss_pvalue
##   <dbl>    <dbl>
## 1     1.49      0.01
```

Table 1. KPSS test of the log transformed Total Fertility rate in Singapore from 1960-1924. The p-value is 0.01, which is significant at the $\alpha = 0.05$ significance level.

```
singapore |> features(log_TLB, unitroot_kpss)
```

```
## # A tibble: 1 x 2
##   kpss_stat kpss_pvalue
##   <dbl>    <dbl>
## 1     0.996      0.01
```

Table 2. KPSS test of the log transformed Total Live-Birth rate in Singapore from 1960-1924. The p-value is 0.01, which is significant at the $\alpha = 0.05$ significance level.

Figures 3 and 4, along with the p-values from tables 1 and 2 both being significant at the 0.05 significance level, show that a log transformation seems to help a bit but they both still need to be stabilized more. To do this, they will be differenced.

```
TFR <- difference(log(singapore$`Total Fertility Rate (TFR)`), 1)
```

```
TLB <- difference(log(singapore$`Total Live-Births`), 1)
```

```
singapore |> autoplot(TFR) +
  labs(title = "First-Order difference of Log TFR (1960-2024)",
        caption = stringr::str_wrap("Figure 5. Plot of the first-order difference of the log transformed",
        y = "Change in log(TFR)",
        x = "Year") +
  theme(plot.caption = element_text(hjust = 0.5),
        plot.title = element_text(hjust = 0.5))
```

```
## Ignoring unknown labels:
```

```
## * width : "80"
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
```

```
## ('geom_line()').
```

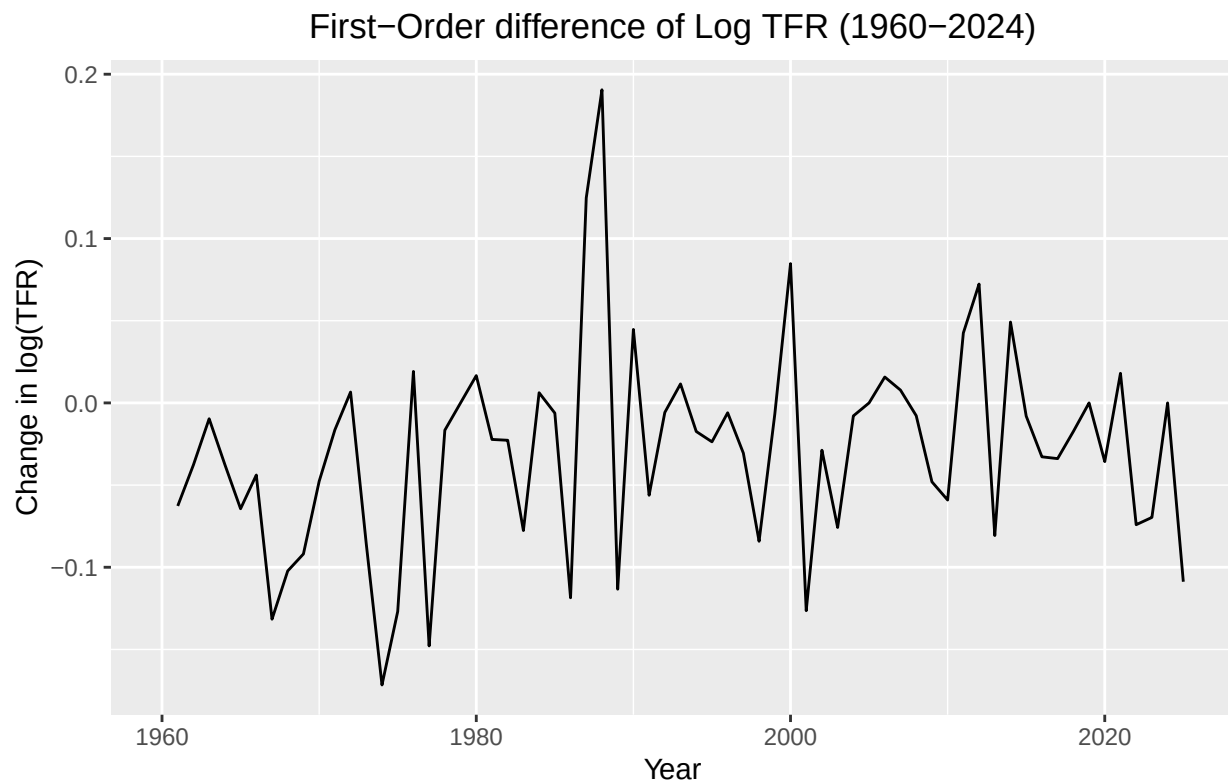


Figure 5. Plot of the first-order difference of the log transformed Total Fertility Rate in Singapore from 1960 to 2024.

```
singapore |> autoplot(TLB) +
  labs(title = "First-Order difference of Log TLB (1960-2024)",
        caption = stringr::str_wrap("Figure 6. Plot of the first-order difference of the log transformed",
        width = 80,
        y = "Change of log(TLB)",
        x = "Year") +
```

```
theme(plot.caption = element_text(hjust = 0.5),
      plot.title = element_text(hjust = 0.5))
```

```
## Ignoring unknown labels:
## * width : "80"
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## ('geom_line()').
```

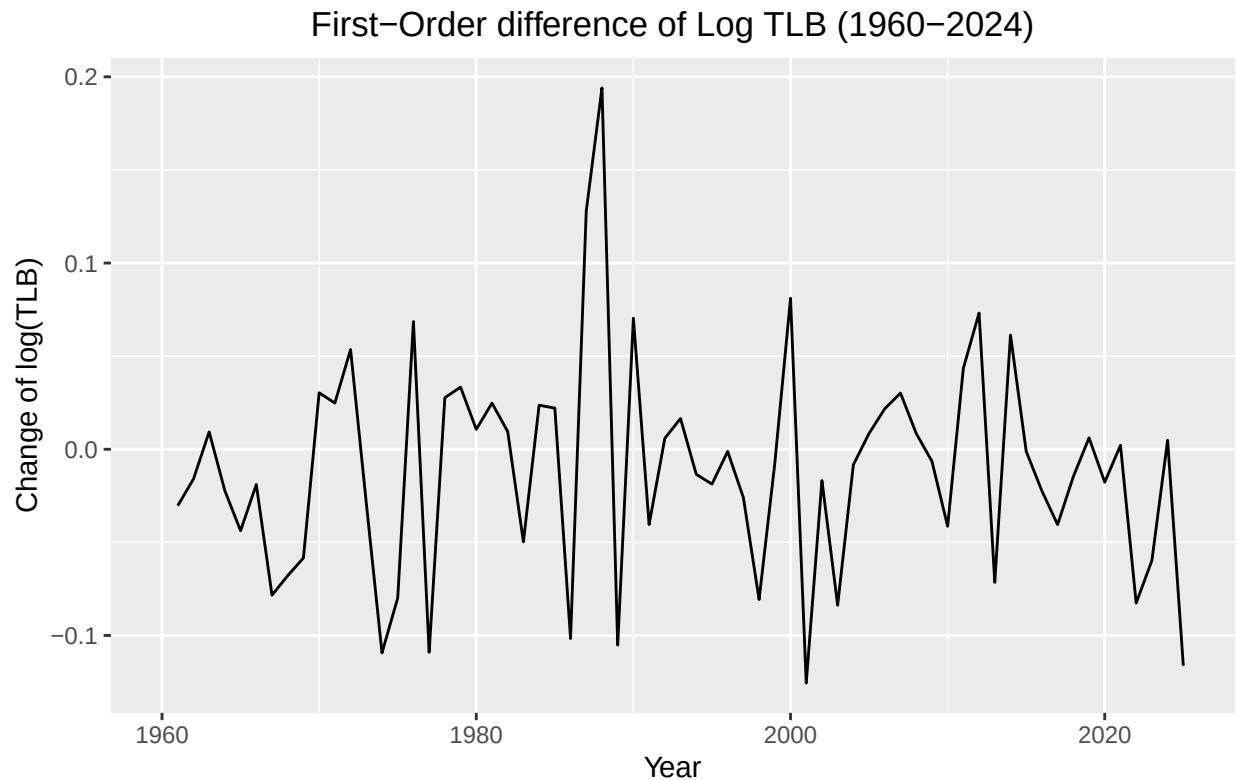


Figure 6. Plot of the first-order difference of the log transformed Total Live-Births in Singapore from 1960 to 2024.

Plots 5 and 6 no longer show a downward trend with a first-order difference of the log transform of each variable.

```
singapore |> features(TFR, unitroot_kpss)
```

```
## # A tibble: 1 x 2
##   kpss_stat kpss_pvalue
##   <dbl>      <dbl>
## 1     0.370      0.0902
```

Table 3. KPSS test of the first-order differenced log transformed Total Fertility rate in Singapore from 1960-1924. The p-value is 0.09, which is not significant at the $\alpha = 0.05$ significance level.

```
singapore |> features(TLB, unitroot_kpss)
```

```
## # A tibble: 1 x 2
##   kpss_stat kpss_pvalue
##   <dbl>     <dbl>
## 1      0.114         0.1
```

Table 4. KPSS test of the first-order differenced log transformed Total Live-Birth rate in Singapore from 1960-1924. The p-value is 0.1, which is not significant at the $\alpha = 0.05$ significance level.

Tables 3 and 4 confirm what figures 5 and 6 showed, when both Total Live-Births and Total Fertility rate are log transformed and then differenced, they are stationary. In a KPSS test, if the p-value is less than 0.05, the null-hypothesis that the series is stationary, is rejected. However, if the p-value is greater than 0.05, the null-hypothesis is accepted and the series is found to be stationary.

```
singapore |> ACF(TFR) |>
  autoplot() +
  labs(title = "ACF of First-Order Differenced Log TFR (1960-2024)",
       caption = stringr::str_wrap("Figure 7. Autocorrelation Function of the stationary Total Fertility",
                                   width = 80)
  ) +
  theme(plot.caption = element_text(hjust = 0.5),
        plot.title = element_text(hjust = 0.5)
  )
```

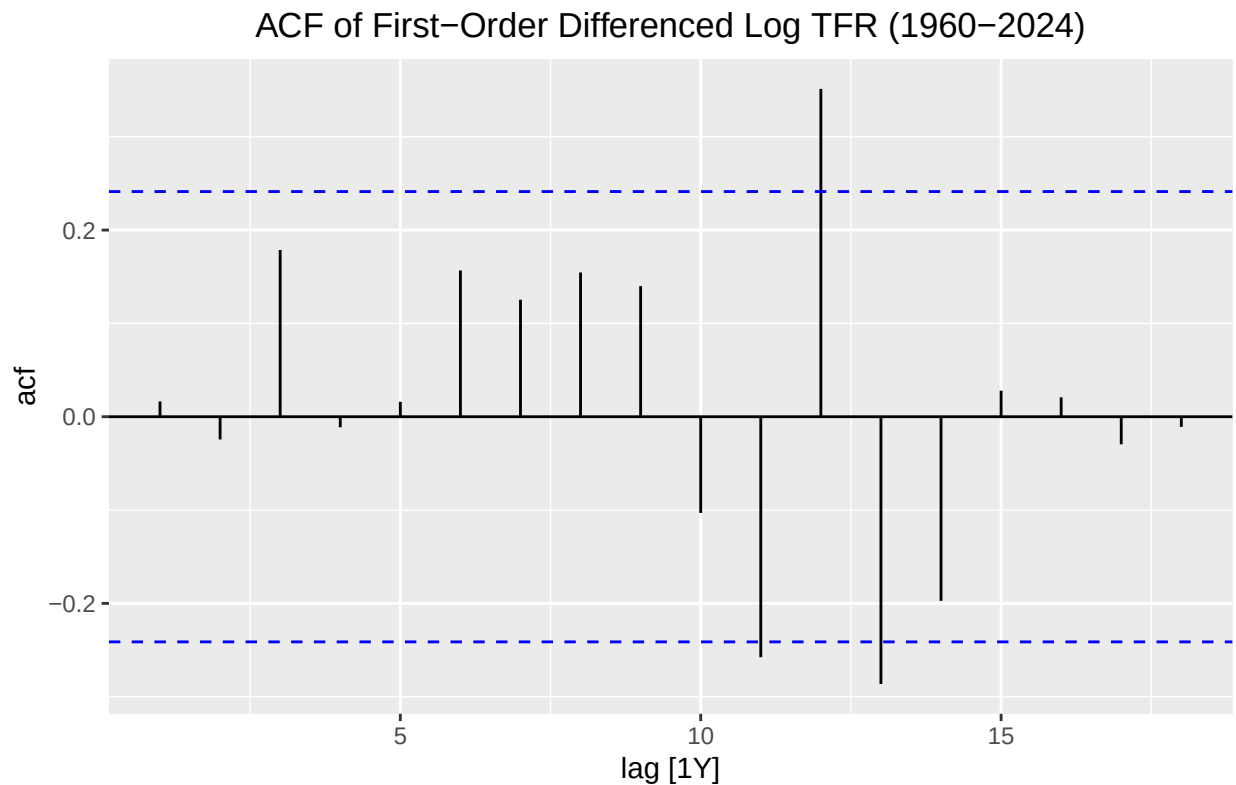


Figure 7. Autocorrelation Function of the stationary Total Fertility Rate data in Singapore from 1960-2024.


```
singapore |> ACF(TLB) |>
  autoplot() +
  labs(title = "ACF of First-Order Differenced Log TLB (1960–2024)",
        caption = stringr::str_wrap("Figure 8. Autocorrelation Function of the stationary Total Live-Bir
  ) +
  theme(plot.caption = element_text(hjust = 0.5),
        plot.title = element_text(hjust = 0.5)
  )
```

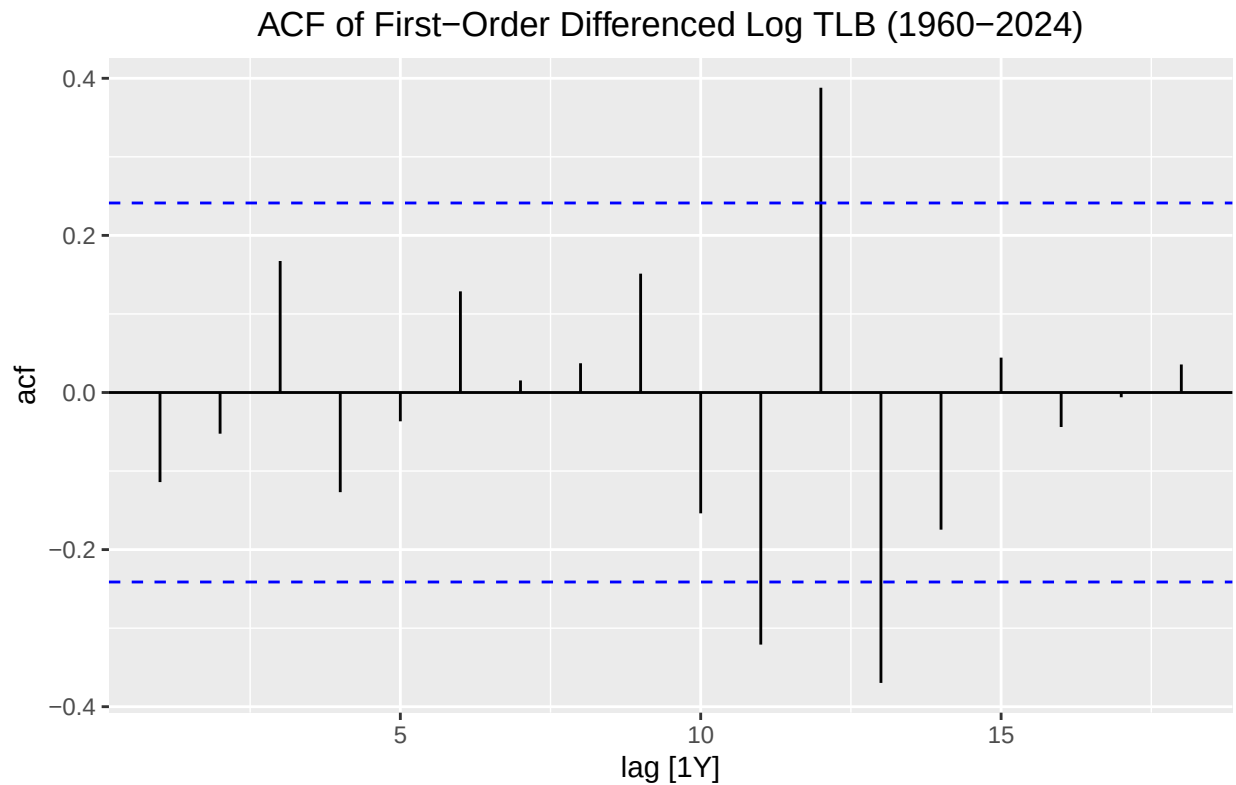


Figure 8. Autocorrelation Function of the stationary Total Live-Births data in Singapore from 1960–2024.

```
singapore |> PACF(TFR) |>
  autoplot() +
  labs(title = "PACF of First-Order Differenced Log TFR (1960–2024)",
        caption = stringr::str_wrap("Figure 9. Partial Autocorrelation Function of the stationary Total L
  ) +
  theme(plot.caption = element_text(hjust = 0.5),
        plot.title = element_text(hjust = 0.5)
  )
```

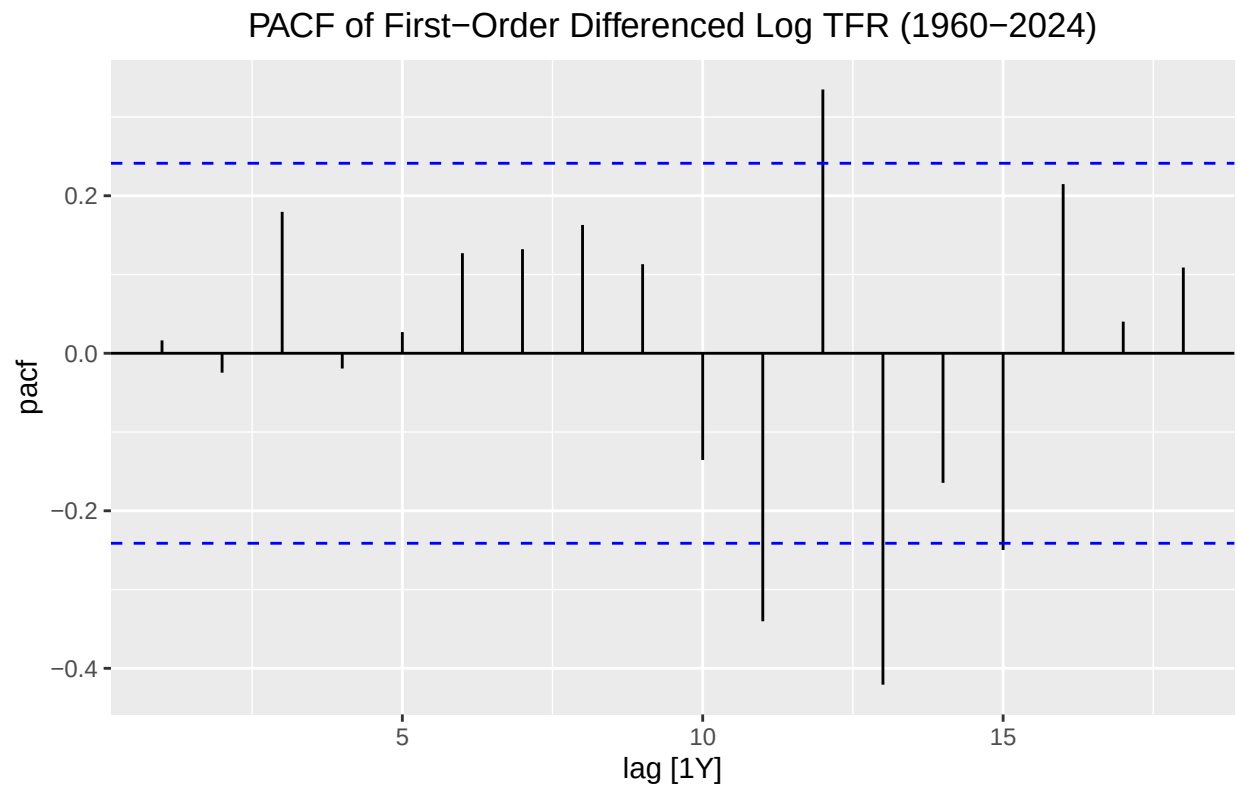


Figure 9. Partial Autocorrelation Function of the stationary Total Fertility Rate data in Singapore from 1960–2024.

```
singapore |> PACF(TLB) |>
  autoplot() +
  labs(title = "PACF of First-Order Differenced Log TLB (1960-2024)",
        caption = stringr::str_wrap("Figure 10. Partial Autocorrelation Function of the stationary Total
  ) +
  theme(plot.caption = element_text(hjust = 0.5),
        plot.title = element_text(hjust = 0.5)
  )
```

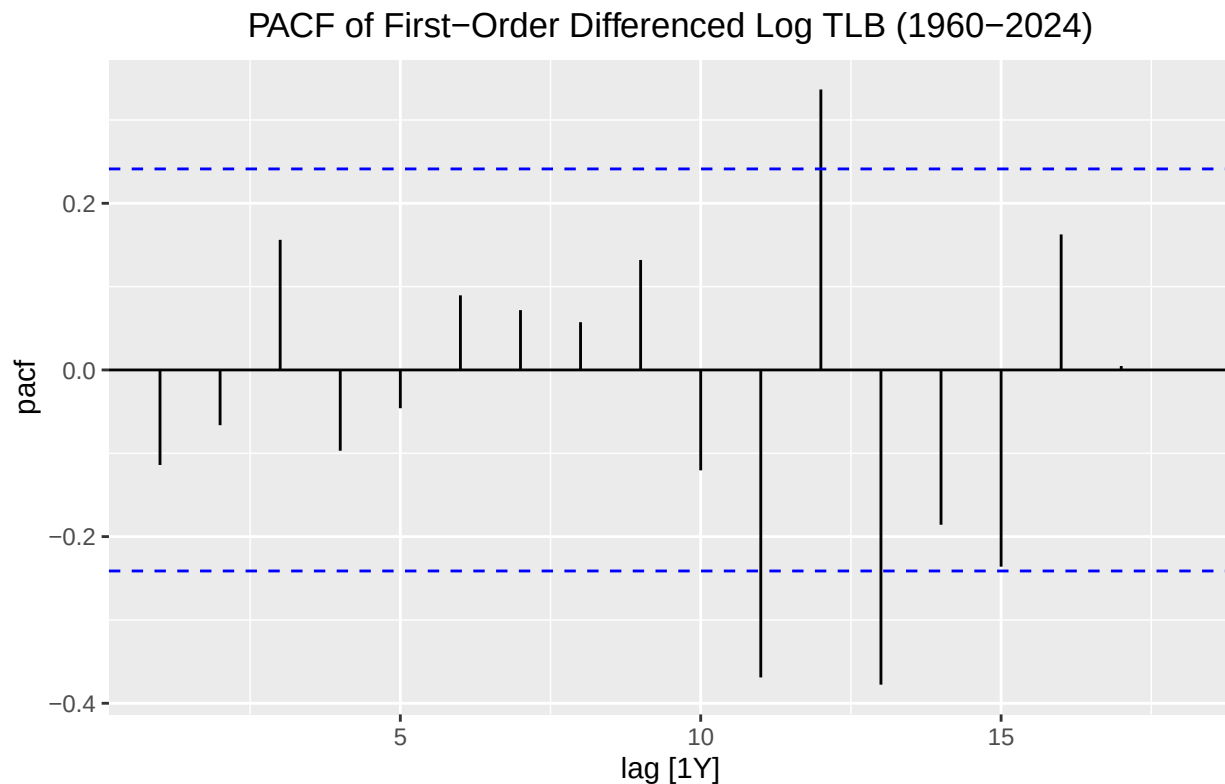


Figure 10. Partial Autocorrelation Function of the stationary Total Live-Births data in Singapore from 1960–2024.

The spikes at lags 11,12,13 in figures 7,8,9, and 10 show that there’s a cyclic behaviour. The year of the Dragon historically has had an effect on birth rates in Singapore. Culturally, it is “considered auspicious for births” (The Straits Times). The Year of the Dragon comes every 12 years, which lines up perfectly with the spikes.

```
TFR_Model1 <- singapore |>
  model(
    tfr_baseline = ARIMA(log_TFR ~ pdq(0,1,0)),
    tfr_auto = ARIMA(log_TFR))
TFR_Model1 |> pivot_longer(everything(), names_to = "Model name", values_to = "Orders")
```

```
## # A mable: 2 x 2
## # Key:   Model name [2]
##   'Model name'      Orders
##   <chr>             <model>
## 1 tfr_baseline <ARIMA(0,1,0) w/ drift>
## 2 tfr_auto    <ARIMA(0,1,0) w/ drift>
```

Table 5. This comparison table shows that the auto ARIMA model chose for the Total Fertility Rate is the same as the chosen ARIMA model: ARIMA(0,1,0).

The ARIMA model is chosen with this in mind: ARIMA(p,d,q). Where p is the value chosen by looking at immediate lags that spike on the PACF, d is chosen by the differencing, and q is chosen by looking at immediate lags that spike on the ACF.

```
TFR_Model1 |> report()
```

```
## # A tibble: 2 x 8
##   .model      sigma2 log_lik   AIC   AICc   BIC ar_roots  ma_roots
##   <chr>      <dbl>   <dbl> <dbl> <dbl> <dbl> <list>   <list>
## 1 tfr_baseline 0.00392   88.4 -173. -173. -168. <cpl [0]> <cpl [0]>
## 2 tfr_auto     0.00392   88.4 -173. -173. -168. <cpl [0]> <cpl [0]>
```

Table 6. This comparison table of the chosen TFR baseline ARIMA model and the automatically chose TFR ARIMA model confirms that they are identical.

```
TFR_Model1 |>
  select(tfr_auto) |> gg_tsresiduals() +
  labs(title = "1960-2024 TFR ARIMA (0,1,0) Plots") +
  theme(plot.title = element_text(hjust = 0.5))
  )
```

```
## Warning: 'gg_tsresiduals()' was deprecated in feasts 0.4.2.
## i Please use 'ggtime::gg_tsresiduals()' instead.
## i Graphics functions have been moved to the {ggtime} package. Please use
##   'library(ggtime)' instead.
## This warning is displayed once per session.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

1960–2024 TFR ARIMA (0,1,0) Plots

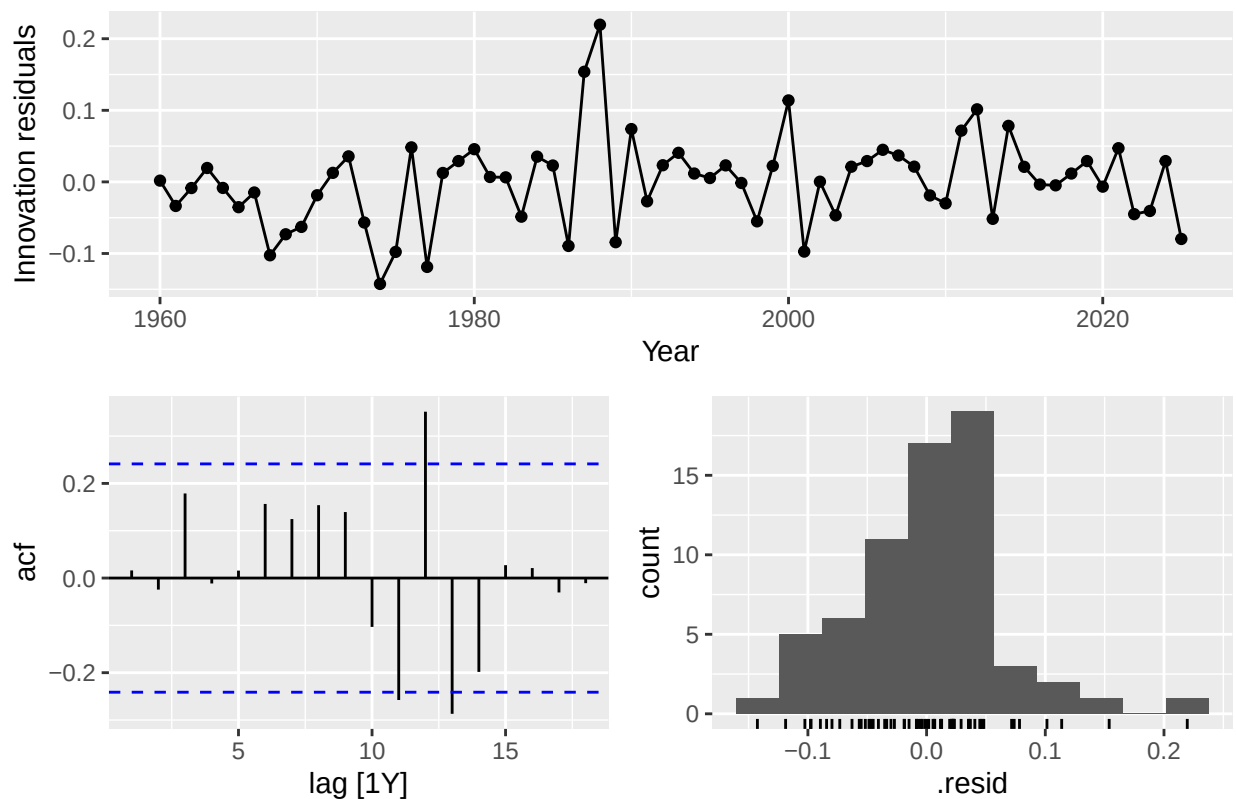


Figure 11. Plots showing the ARIMA(0,1,0) model for Total Fertility Rate in Singapore from 1960-2024. The residuals plot shows the errors bounding roughly around zero. The ACF plot shows spikes at lags 11, 12, and 13. The histogram is unimodal with an abrupt drop off around 0.05 and an outlier around 0.2.

```
TLB_Model <- singapore |>
  model(
    tlb_baseline = ARIMA(log_TLB ~ pdq(0,1,0)),
    tlb_auto = ARIMA(log_TLB))
TLB_Model |> pivot_longer(everything(), names_to = "Model name", values_to = "Orders")
```

```
## # A mable: 2 x 2
## # Key:      Model name [2]
##   'Model name'      Orders
##   <chr>             <model>
## 1 tlb_baseline <ARIMA(0,1,0) w/ drift>
## 2 tlb_auto     <ARIMA(0,1,0) w/ drift>
```

Table 7. This comparison table shows that the auto ARIMA model chose for the Total Live-Births is the same as the chosen ARIMA model: ARIMA(0,1,0).

```
TLB_Model |> report()
```

```
## # A tibble: 2 x 8
##   .model      sigma2 log_lik   AIC  AICc   BIC ar_roots  ma_roots
##   <chr>      <dbl>   <dbl> <dbl> <dbl> <dbl> <list>   <list>
## 1 tlb_baseline 0.00348   92.3 -181. -180. -176. <cpl [0]> <cpl [0]>
## 2 tlb_auto     0.00348   92.3 -181. -180. -176. <cpl [0]> <cpl [0]>
```

Table 8. This comparison table of the chosen TLB baseline ARIMA model and the automatically chose TLB ARIMA model confirms that they are identical.

```
TLB_Model |> select(tlb_auto) |> gg_tsresiduals() +
  labs(title = "1960-2024 TLB ARIMA (0,1,0) Plots")
```

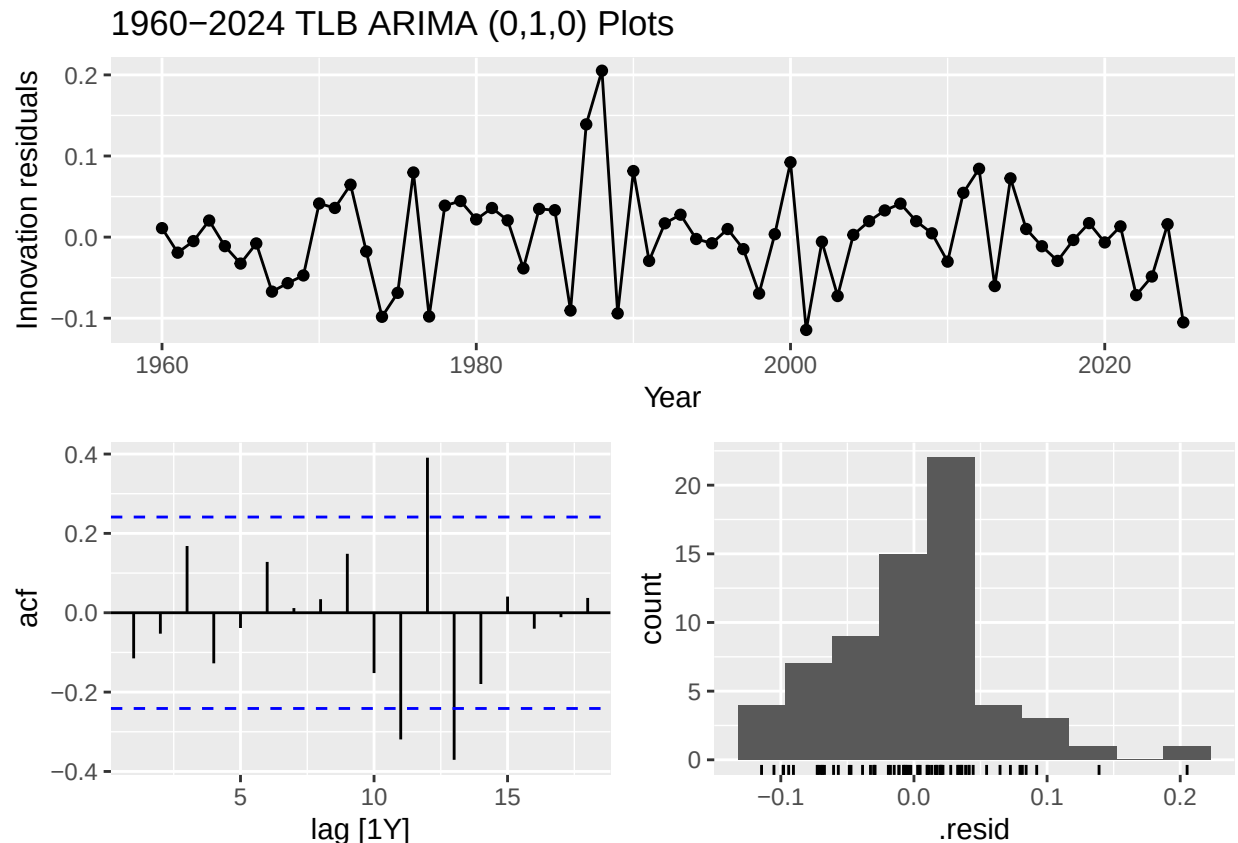


Figure 12. Plots showing the ARIMA(0,1,0) model for Total Live Births in Singapore from 1960-2024. The residuals plot shows the errors bounding roughly around zero. The ACF plot shows spikes at lags 11, 12, and 13. The histogram is unimodal with an outlier around 0.2.

```
TFR_Train <- singapore |>
  filter(Year <= 2012)
```

```
TFR_train_model1 <- TFR_Train |>
  model(tfr_train_baseline = ARIMA(log_TFR ~ pdq(0,1,0)))
TFR_train_forecast <- TFR_train_model1 |>
  forecast(h=12)
TFR_train_forecast |> autoplot() +
  autolayer(singapore, `Total Fertility Rate (TFR)`) +
  labs(
    title = "ARIMA(0,1,0) Predicted vs Actual TFR",
    caption = stringr::str_wrap(
      "Figure 13. Plot shows that the predictive model drops significantly more than the actual data do
    )
  )
```

```
## Warning: 'autoplot.fbl_ts()' was deprecated in fabletools 0.6.0.
## i Please use 'ggtime::autoplot.fbl_ts()' instead.
## i Graphics functions have been moved to the {ggtime} package. Please use
##   'library(ggtime)' instead.
## This warning is displayed once per session.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
```

```
## generated.

## Warning: 'autolayer.tbl_ts()' was deprecated in fabletools 0.6.0.
## i Please use 'ggtime::autolayer.tbl_ts()' instead.
## i Graphics functions have been moved to the {ggtime} package. Please use
## 'library(ggtime)' instead.
## This warning is displayed once per session.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

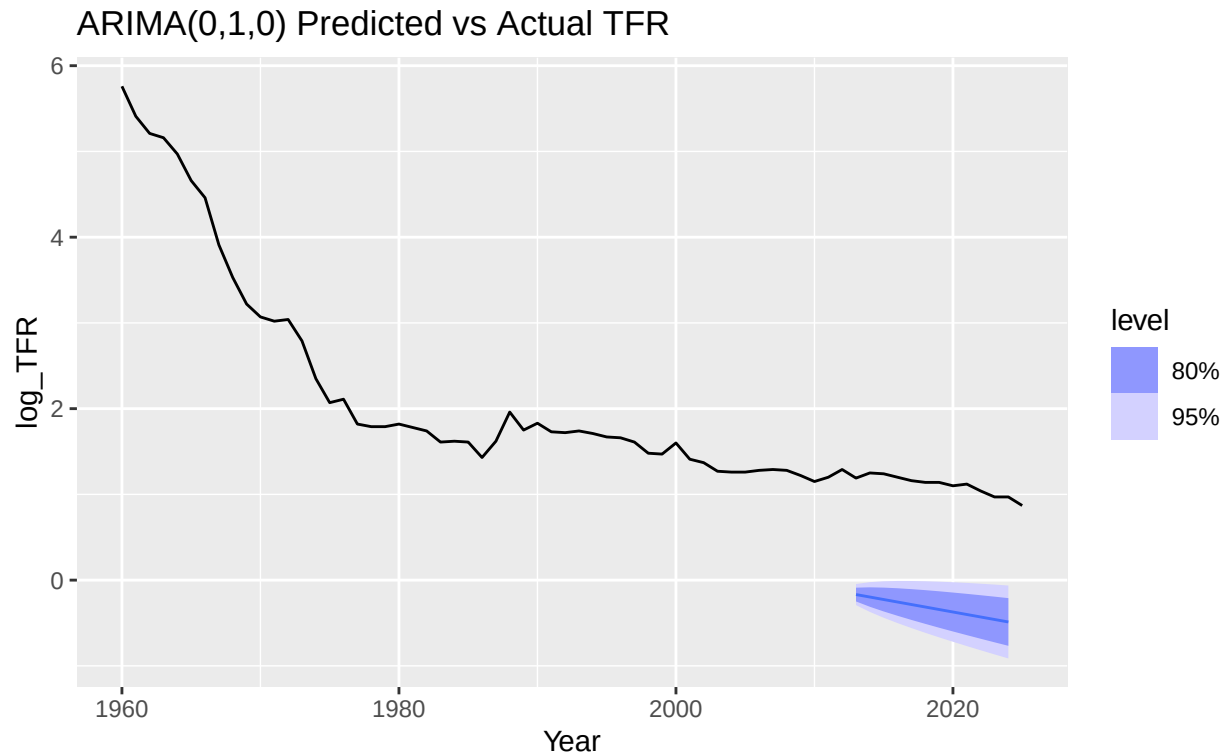


Figure 13. Plot shows that the predictive model drops significantly more than the actual data does for the 2013–2024 period for the Total Fertility Rate in Singapore.

```
TLB_Train <- singapore |>
  filter(Year <= 2012)
```

```
TLB_train_model1 <- TLB_Train |>
  model(tlb_train_baseline = ARIMA(log_TLB ~ pdq(0,1,0)))
TLB_train_forecast <- TLB_train_model1 |>
  forecast(h=12)
TLB_train_forecast |> autoplot() +
  autolayer(singapore, `Total Live-Births`) +
  labs(
    title = "ARIMA(0,1,0) Predicted vs Actual TLB",
    caption = stringr::str_wrap(
      "Figure 14. Plot shows that the predictive model drops significantly more than the actual data does"
    )
  )
```

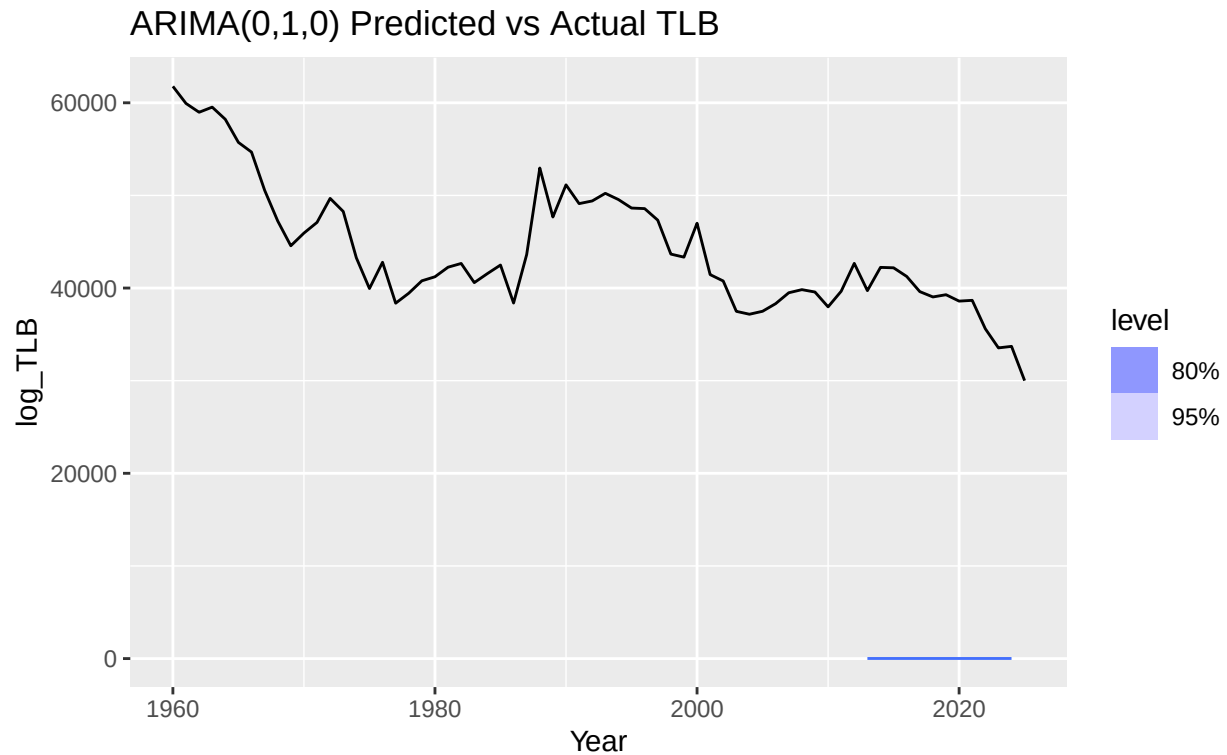


Figure 14. Plot shows that the predictive model drops significantly more than the actual data does for the 2013–2024 period for the Total Live Births in Singapore.

Figures 13 and 14 show that an ARIMA model is not adequate enough at making a predictive modeling the Total Live Births and Total Fertility Rate from 2013–2024 base on the data from 1960–2012.

```
TFR_train_model2 <- TFR_Train |>
  model(tfr_linear = TSLM(log(`Total Fertility Rate (TFR)`) ~ trend()))
TFR_train_forecast2 <- TFR_train_model2 |>
  forecast(h=12)
TFR_train_forecast2 |> autoplot() +
  autolayer(singapore, `Total Fertility Rate (TFR)`) +
  labs(
    title = "TSLM Predicted vs Actual TFR",
    caption = stringr::str_wrap(
      "Figure 15. Plot shows that the predictive model drops slightly more than the actual data does for
    )
  )
```

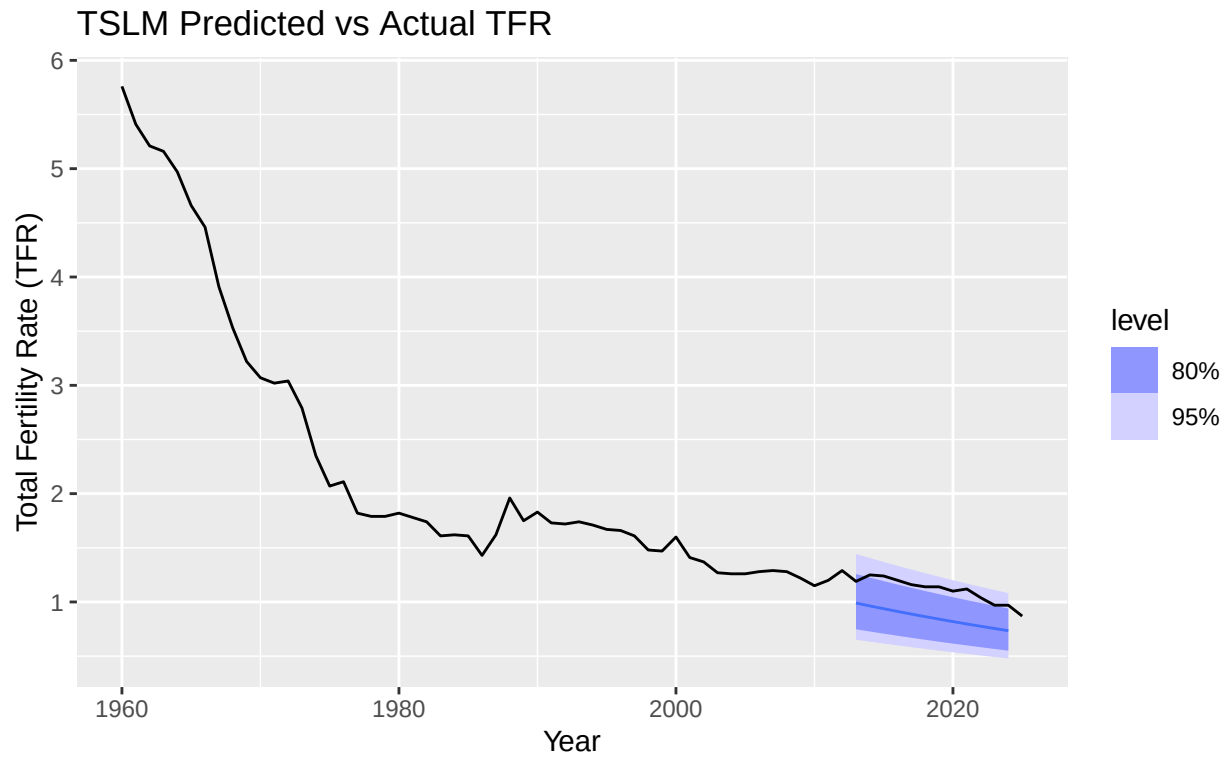



Figure 15. Plot shows that the predictive model drops slightly more than the actual data does for the 2013–2024 period for the Total Live Births in Singapore.

```
TLB_train_model2 <- TLB_Train |>
  model(tlm_linear = TSLM(log(`Total Live-Births`) ~ trend()))
TLB_train_forecast2 <- TLB_train_model2 |>
  forecast(h=12)
TLB_train_forecast2 |> autoplot() +
  autolayer(singapore, `Total Live-Births`) +
  labs(
    title = "TSLM Predicted vs Actual TLB",
    caption = stringr::str_wrap(
      "Figure 16. Plot shows that the predictive model does not drops as significantly as the actual data"
    )
  )
```

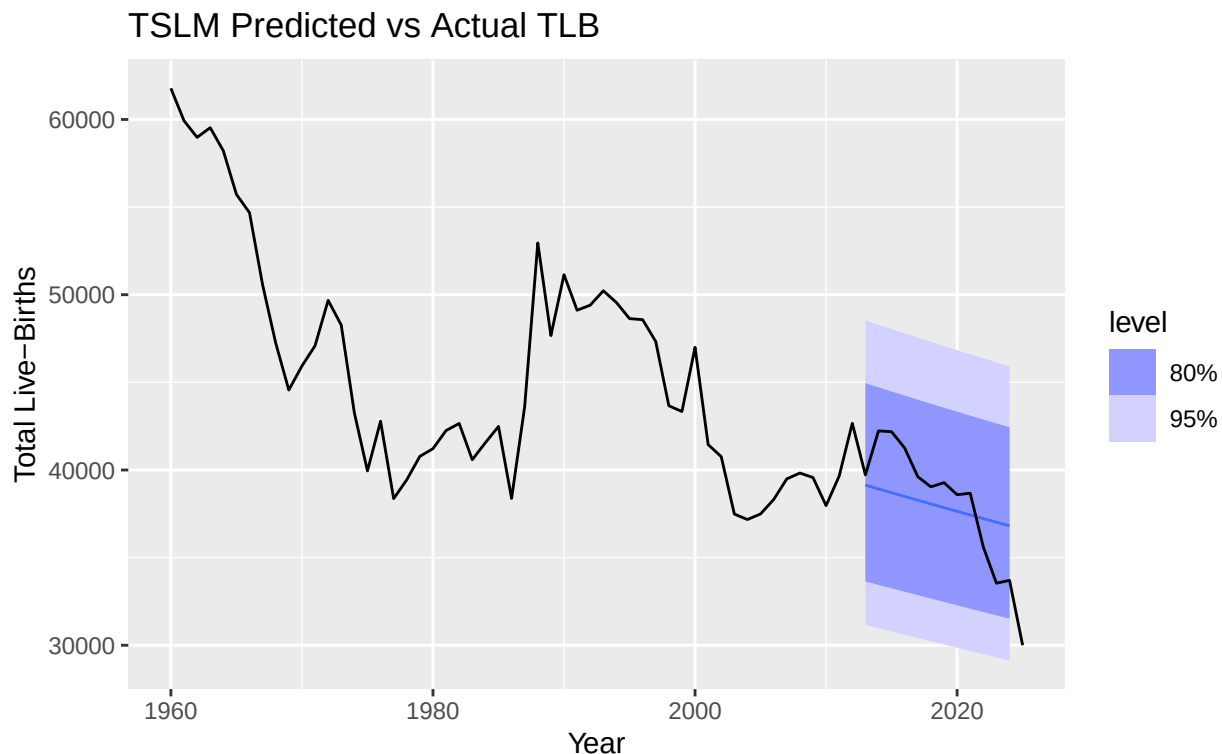


Figure 16. Plot shows that the predictive model does not drops as significantly as the actual data does for the 2013–2024 period for the Total Live Births in Singapore.

```
TFR_Model12 <- singapore |>
  model(
    tfr_baseline = ARIMA(log_TFR ~ pdq(0,1,0)),
    tfr_sarima = ARIMA(log_TFR ~ pdq() + PDQ(period = 12))
  )
TFR_Model12
```

```
## # A mable: 1 x 2
##           tfr_baseline           tfr_sarima
##           <model>             <model>
## 1 <ARIMA(0,1,0) w/ drift> <ARIMA(0,1,1)(1,0,1)[12] w/ drift>
```

Table 9. Shows the automated search selects a SARIMA(0,1,1)(1,0,1)[12] model with drift for the Total Fertility Rate data.

```
glance(TFR_Model12)
```

```
## # A tibble: 2 x 8
##   .model      sigma2 log_lik   AIC   AICc   BIC ar_roots  ma_roots
##   <chr>      <dbl>  <dbl> <dbl> <dbl> <dbl> <list>   <list>
## 1 tfr_baseline 0.00392   88.4 -173. -173. -168. <cpl [0]> <cpl [0]>
## 2 tfr_sarima  0.00292   96.7 -183. -182. -172. <cpl [12]> <cpl [13]>
```

Table 10. Shows the SARIMA(0,1,1)(1,0,1)[12] model for Total Fertility Rate has a lower AIC at around -183, nearly 10 points lower when compared to the ARIMA(0,1,0) model for Total Fertility Rate. The BIC is also lower for the SARIMA model, and the sigma² value is also significantly lower for the SARIMA model.

All of the values in table 10 confirm that a seasonal ARIMA model is better at handling the Total Fertility Rate data in Singapore from 1960-2024.

```
TFR_train_sarima <- TFR_Train |>  
  model(TFR_train_sarima = ARIMA(log_TFR ~ pdq(0,1,1) + PDQ(1,0,1, period = 12)))
```

```
TFR_train_sarima_fc <- TFR_train_sarima |>  
  forecast(h = 12)
```

```
TFR_train_sarima_fc |>  
  autoplot() +  
  autolayer(singapore, `Total Fertility Rate (TFR)`)
```

